

Configurable Enterprise Workflow Automation

**Company Name:
RAMRAJ COTTON**

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INTERNSHIP DETAILS



- Name:Harita S
- Internship position: software developer
- Organization: Ramraj Cotton
- Mode of internship: Offline
- Start & End Date of internship: 18.05.2026 – 31.10.2026
- Guide: Varunan R [IT-GM]
- type of project:AI-Based Web Application Development (Intelligent Document Processing & Workflow Automation)

PROJECT INTRODUCTION



TITLE : Configurable Enterprise Workflow Automation

- AI-powered web application for intelligent document processing and configurable workflow automation.
- Provides a dynamic workflow builder that allows administrators to create and modify approval workflows without code.
- Uses OCR to extract information from documents and LLMs to interpret and validate the extracted data.
- Integrates n8n to automate email monitoring, document intake, and workflow initiation.
- Enables secure document management, multi-level approvals, status tracking, and role-based access control.



Problem Definition or Objectives or TASKS

Objectives

The primary objective of this project is to develop an AI-powered web application that automates document processing and approval workflows. The system is designed to reduce manual effort, improve processing efficiency, and provide configurable workflows that can adapt to changing organizational requirements.

The specific objectives are:

- Develop an AI-powered enterprise web application for document management.
- Automate document processing using OCR and Large Language Models (LLMs).
- Implement configurable approval workflows that allow administrators to create and modify workflows without coding.
- Integrate n8n for email monitoring, document intake, and workflow automation.
- Improve processing accuracy, reduce turnaround time, and enhance overall operational efficiency.

Tools or Technology Used



Software & Frameworks

- React.js, Node.js, Express.js
- Microsoft SQL Server
- PaddleOCR, Ollama (LLM)
- n8n, Docker

Programming Languages

- Python
- JavaScript
- SQL
- HTML, CSS

Development Tools

- Visual Studio Code
- Postman
- Git & GitHub
- Docker Desktop

Hardware & Platform

- Development: Windows 11
- Deployment: Ubuntu Server
- Processor: Intel Core i5
- Memory: 16 GB RAM



Functional Requirements

- User Authentication & Role-Based Access
- Document Upload & Management
- OCR & AI-Based Document Processing
- Configurable Workflow Management
- Multi-Level Approval Process
- Email Notifications & Workflow Automation
- Dashboard, Search & Status Tracking
- Reports & Audit Logs



Non-Functional Requirements

- Security and Role-Based Access Control
- Responsive User Interface
- High Performance and Reliability
- Scalability and Maintainability
- Data Integrity and Secure Storage

Use CASE Design

Use Case: Email to Automated Workflow Processing





Test Plan

- Email Integration – Verify email and attachment import.
- Document Processing – Validate OCR and AI data extraction.
- Workflow Execution – Verify configurable workflow routing.
- Approval Process – Validate approval, rejection, and notifications.
- System Integration – Test the complete end-to-end workflow.

Testing Types

- Functional Testing
- Integration Testing
- System Testing
- User Acceptance Testing (UAT)

Test plan



Test Module	Objective	Testing Method	Expected Result
User Authentication	Verify secure login and role-based access	Functional Testing	Authorized users can log in successfully
Email Integration	Verify email monitoring and attachment import using n8n	Integration Testing	Documents are automatically received and imported
Document Upload	Validate document upload and storage	Functional Testing	Documents are uploaded and stored correctly
OCR & AI Processing	Verify text extraction and AI-based data validation	Functional & Integration Testing	Accurate document data extraction
Workflow Execution	Validate configurable workflow routing	Integration Testing	Documents follow the configured approval flow
Approval Process	Verify approval, rejection, and status updates	Functional Testing	Workflow status updates correctly
Notifications	Validate email/system notifications	Functional Testing	Notifications are delivered successfully
System Integration	Verify complete end-to-end workflow	System Testing	Entire application functions correctly

Work completed till date and future plan of action



Work Completed Till Date

- Completed requirement analysis and system architecture design.
- Developed core backend modules, database, and authentication.
- Implemented initial OCR, AI (LLM), and n8n integration.
- Developed frontend modules including Login, Dashboard, and Workflow Builder.
- Performed module integration, testing, and issue resolution.

Future Plan of Action

- Complete the remaining workflow and approval modules.
- Enhance OCR and AI extraction accuracy.
- Complete end-to-end system integration and testing.
- Optimize application performance and resolve pending issues.
- Finalize documentation and prepare for deployment.

Conclusion



- Successfully completed the project planning, system design, and core module development.
- Implemented the initial integration of OCR, AI (LLM), and n8n for document processing and workflow automation.
- Gained practical experience in full-stack development, AI integration, and enterprise workflow automation.
- The remaining development, testing, optimization, and deployment will be completed in the next phase of the internship.